

SHUBHASHIS SINHA

Staff Engineer — Databricks & AI Systems | GenAI Delivery · Client Advisory · Prototype-to-Production

✉ shubhashis.sinha@gmail.com | 📞 +91-7358604077 | 📍 Bangalore, India | 🔗 linkedin.com/in/shubhashis-sinha

SUMMARY

Engineer with **10+ years building and shipping AI-powered data systems**, the last 3+ years focused on **GenAI delivery on Databricks and AWS** — from ambiguous business asks to production-grade solutions. Proven track record of **shaping use cases with business stakeholders**, building working prototypes in weeks, and productionising them with CI/CD, monitoring, and documentation. Sole technical SPOC for cross-functional business teams — advising on what to build, what not to build, and managing trade-offs under ambiguity. Active user of **AI-assisted engineering (Claude, Codex)** to accelerate design, coding, testing, and documentation. Built reusable frameworks and accelerators adopted across 6+ teams. Delivered **\$2M+ cost savings**, 10TB+/day platform throughput, and GenAI systems serving 1,000+ requests/month.

CORE COMPETENCIES

Databricks & AI	Databricks (Unity Catalog, Genie, Vector Search, MLflow, AI Gateway, AI/BI Dashboards) · Databricks Workflows · Delta Lake · Feature Store · Model Serving · FinOps on Databricks
GenAI / LLM Systems	Production RAG · Agentic Workflows (LangGraph) · LLMops (MLflow, LangSmith) · Evaluation Frameworks · Prompt Management · Human-in-the-Loop Controls · Model Risk Basics · Amazon Bedrock (Claude)
AI Tool Usage	Claude (Anthropic) · Codex (OpenAI) · AI-assisted design, code generation, testing, debugging, and documentation — used daily to accelerate delivery velocity
Data Engineering	Spark (PySpark/Scala) · Kafka/MSK · AWS Glue · EMR · Redshift · Athena · Snowflake · Apache Iceberg · Real-Time & Batch Pipelines
Software Engineering	Production-grade APIs (FastAPI) · CI/CD · Testing (unit, integration, evaluation) · Logging & Monitoring · Versioning · Docker · Kubernetes/EKS · Terraform
Client Advisory	Use-case shaping · Trade-off communication · Stakeholder management · Ambiguity handling · Brief redirection · Working sessions facilitation · Technical storytelling for business audiences
IP & Accelerators	Reusable frameworks · Prompt libraries · Reference implementations · Delivery playbooks · Runbooks · Pattern extraction from live delivery

PROFESSIONAL EXPERIENCE

Staff Engineer — AI & Data Platform (Technical SPOC)

Dec 2021 – Present

GE HealthCare · Bangalore, India

Databricks-Native GenAI Delivery & Agentic AI

- Deployed **GraphRAG platform on Databricks** — leveraged **Unity Catalog** for governance, **Vector Search** for embedding retrieval, and **MLflow** for experiment tracking, model versioning, and evaluation. LangGraph-orchestrated agents with template routing, multi-step reasoning, and NL-to-SPARQL fallback. **Prototype to production in 8 weeks.**
- Built **LLMops pipeline on Databricks** — **AI Gateway** for model routing and rate limiting, **MLflow** for prompt versioning and A/B evaluation, **Genie** for natural-language data exploration by business users. Established evaluation frameworks with human-in-the-loop review gates.
- Shipped **Contract Digitalization using Agentic AI** — autonomous agents processing **1,000+ contracts/month**. Designed human-in-the-loop controls for confidence thresholds, escalation routing, and audit trails. Owned full lifecycle: use-case shaping → architecture → production monitoring.
- Built **GenAI-driven Rally automation** — RAG retrieval + LLM orchestration (Bedrock Claude 3.5 + Step Functions). End-to-end: requirement → story generation → task decomposition → auto-assignment → HLD/LLD/pseudocode. **Reduced turnaround from ~2 days to hours (~40% effort reduction)**, adopted by 4+ teams / 20+ developers.
- Shipped **production RAG system** (Bedrock Claude 3.5 + pgvector) — intelligent search across **10,000+ clinical documents**, sub-second P95 latency. Used **Claude and Codex** to accelerate prompt engineering, retrieval tuning, and test generation.

Client Advisory, Use-Case Shaping & Stakeholder Management

- Sole technical SPOC** for 3 business verticals (Supply Chain, Clinical, Finance) — converted vague AI ambitions into clear, buildable use cases. Led discovery workshops, defined scope, and advised on sequencing and trade-offs.
- Redirected briefs** on multiple occasions — advised business leadership on what *not* to build (e.g., rejected a full knowledge-graph rebuild in favour of targeted GraphRAG overlay, saving 4 months of effort). Managed pushback and maintained stakeholder confidence through data-driven trade-off communication.
- Operated as **embedded technical advisor** for cross-company vendor engagements — shaped direction with external partners (Neptune DB vendor, Databricks solution architects), managed skeptical audiences during POV evaluations, and defended architectural decisions under scrutiny.

- Led **working sessions with business and technical stakeholders** — translated complex AI/ML trade-offs into plain language, set realistic expectations, and maintained confidence through ambiguous, high-pressure delivery phases.
- Managed **Databricks vendor relationship** — evaluated Genie, AI/BI, and Vector Search capabilities during early-access phases, provided structured feedback to Databricks solution architects, and shaped internal adoption strategy.

Reusable IP, Frameworks & Accelerators

- Created **reusable PySpark framework** adopted by 6+ teams — standardized ingestion, transformation, and quality patterns. **25% faster pipeline development** across the platform.
- Built **RAG accelerator kit** — prompt libraries, evaluation harnesses, retrieval templates, and deployment runbooks. Reduced new RAG project bootstrap from weeks to days.
- Authored **delivery playbooks** for GenAI engagements — covering use-case qualification, architecture decision records, production readiness checklists, and LLMOps patterns. Shared across engineering org.
- Extracted **reusable agentic workflow patterns** — configurable agent templates, tool-calling abstractions, and HITL control patterns packaged as internal accelerators.

High-Throughput Data Platform & Production Excellence

- Designed and operated **pipelines processing 10TB+ daily** — 99.9% uptime across ingestion, transformation, and serving layers.
- Led **Teradata-to-Redshift migration** — re-engineered 500+ pipelines, delivering **40% cost reduction (\$2M+ annual savings)** with 3x query performance improvement.
- Built **real-time streaming** (Kafka/MSK + PySpark on EMR) — reduced data latency from hours to **under 5 minutes** for ML feature pipelines.
- Drove **FinOps across Redshift, Databricks, EMR** — right-sizing, spot instances, workload scheduling. Built automated data quality framework (200+ rules), reduced incidents by 35%.

Tech Stack: Databricks (Unity Catalog, Genie, Vector Search, MLflow, AI Gateway, AI/BI, Workflows, Delta Lake) · Python, PySpark, Scala, SQL · AWS (S3, Glue, Athena, Redshift, EMR, EKS, Neptune, SageMaker, Bedrock, Lambda, Step Functions) · LangGraph/LangChain · RAG/Embeddings · pgvector · FastAPI · Docker · Terraform · Claude · Codex

Senior Data Engineer

Bank of America · Chennai, India

Oct 2017 – Dec 2021

- Architected **Spark batch and streaming pipelines** for regulatory reporting — processing **5TB+ daily** with 20% faster transformations on EMR and Snowflake.
- Designed **scalable data models** with incremental loading and full lineage — improved accuracy by 15%, reduced costs by 25%.
- Collaborated with **cross-functional compliance teams** — translated regulatory requirements into technical specifications, managed competing priorities from legal, risk, and technology stakeholders.

Tech Stack: Apache Spark, Python, Scala, SQL, Snowflake, Kafka, AWS (S3, EMR, Glue), Hive

Data Engineer

T-Mobile · Bangalore, India

Mar 2014 – Oct 2017

- Built **data pipelines at scale** (Hadoop, Hive) across **5M+ subscribers** — 25% improvement in data-driven decisions.
- Implemented **CDC/incremental ingestion** — reduced data freshness from T+1 to near real-time for downstream applications.

Tech Stack: Hadoop, Hive, Sqoop, Python, SQL

EDUCATION

Bachelor of Technology — Electronics & Communications Engineering

West Bengal University of Technology · Kolkata, India · GPA: 8.87/10

AWARDS & CERTIFICATIONS

- 🏆 **Sir M. Visvesvaraya Award — GE Healthcare (2024)**
Recognized for shipping production GenAI/RAG systems with measurable business impact.
- 🏆 **Impact Award — GE Healthcare (Oct 2022, Nov 2023, Mar 2024)**
Platform migration (\$2M+ savings), reusable frameworks, AI-driven automation.
- ☁️ **AWS Certified Solutions Architect – Associate (2025)**
- ☁️ **Databricks Certified Data Engineer Associate (2024)**